# LITERATURE REVIEW AND FEATURE REMOVAL

## 1. Correlation-based feature removal

We first performed a correlation analysis on the eligible NHANES cohort. The main highly correlated feature pairs identified were:

- **BMXBMI – BMXWAIST:** r = 0.914
- **DR1TKCAL – DR1TTFAT:** r = 0.885
- **DR1TKCAL – DR1TCARB:** r = 0.878
- **LBXSATSI – LBXSASSI:** r = 0.804

We then ran two experiments using the **same train/test split and XGBoost pipeline**:

**Experiment 1 – Removal of dietary variables**

Removed:
DR1TKCAL, DR1TPROT, DR1TCARB, DR1TTFAT

| Model | MAE | RMSE | $R^2$ |
|---|---|---|---|
| Baseline XGBoost | 35.23 | 44.64 | 0.489 |
| Tuned XGBoost | 34.54 | 43.88 | **0.506** |

**Experiment 2 – Removal of additional highly correlated variables**

In addition to the dietary variables, we removed **BMXBMI** and **LBXSASSI**, retaining **BMXWAIST** and **LBXSATSI**.

| Model | MAE | RMSE | $R^2$ |
|---|---|---|---|
| Baseline XGBoost | 35.42 | 45.05 | 0.479 |
| Tuned XGBoost | 34.89 | 44.24 | **0.498** |

The results suggest that **removing the dietary variables improved the tuned model relative to the corresponding baseline, but additionally removing BMI and AST resulted in a decrease in $R^2$ from 0.506 to 0.498**. Therefore, we don't think correlation alone is sufficient to decide which feature to remove, and we are treating the correlation analysis as a feature-reduction experiment rather than automatically eliminating all highly correlated variables.

**Paper Title:** Hepatic Steatosis Severity Prediction in Nonobese Individuals: Machine Learning Model Development and Validation
Findings from the paper

**Relevant Findings**

1. **Multicollinearity exists among important metabolic predictors.**
2. **Systematic feature selection** (LASSO + RFE) reduced 42 variables to 14.
3. **BMI and waist circumference** were among the strongest predictors.
4. **ALT and AST** were also highly influential.
5. **Metabolic markers** such as triglycerides, TyG and CMetS contributed strongly.
6. **Reducing the feature set caused some performance loss** in their simplified model.
7. **XGBoost performed best** among the six ML models tested.
8. **Nonlinear effects** were observed for important predictors such as BMI, waist, ALT and UA.
9. **Population-specific modeling mattered**: their nonobese-specific model outperformed a general-population model.
10. **10-fold CV + external validation** were used to assess model robustness.

**Potential experiments for HepatoTwin**

1. **Correlation filtering** — Remove highly correlated predictors and check whether reducing redundancy improves CV $R^2$.
2. **LASSO selection** — Use LASSO to identify the most useful predictors, then train XGBoost using only those features.
3. **RFE selection** — Use recursive feature elimination to find a smaller feature set and compare its performance with the current model.
4. **BMI vs waist** — Compare models using BMI, waist circumference, or both to see which anthropometric information contributes more.
5. **Metabolic expansion** — Add relevant metabolic markers available in NHANES and test whether they improve CAP prediction.
6. **Full vs reduced model** — Compare the current full model with a compact model to measure the performance cost of removing features.
7. **Nonlinear features** — Test clinically meaningful ratios/interactions to capture relationships that raw variables may miss.
8. **Subgroup analysis** — Check whether model performance changes across age, sex, or BMI groups.

**Paper Title:** Simple surrogate equations to predict controlled attenuation parameter values for screening non-alcoholic fatty liver disease in a Chinese population

**Findings from the paper**

1. **BMI, triglycerides, neck circumference, and C-peptide were independent CAP predictors.**
2. **BMI and neck circumference had strong correlations with CAP** (r = 0.80 and 0.79).
3. **Several metabolic/liver variables were correlated with CAP**, including ALT, AST, γ-GT, TG, insulin, glucose and HbA1c.
4. **The full four-variable regression model achieved $R^2 = 0.764$** in the derivation cohort.
5. **Removing TG and C-peptide reduced performance**: the simpler anthropometric model had $R^2 = 0.696$.
6. **The model still performed reasonably in validation**, with $R^2 = 0.653$ for CAP1 and 0.625 for CAP2.
7. **The selected predictors had acceptable multicollinearity**, with VIF < 5 for CAP1.
8. **The paper specifically used correlation screening followed by stepwise feature selection** before building the regression model.
9. **Neck circumference was highlighted as an accessible measure of ectopic/visceral fat** and potentially useful for NAFLD screening.
10. **The study was limited by its small, single-center Chinese cohort**, so the authors emphasize validation in other populations.

**Potential experiments for HepatoTwin**

1. **Neck circumference** — Add BMXNECK and test whether it provides additional information about liver fat beyond BMI and waist.
2. **Triglycerides** — Add triglycerides (LBXTR) because the paper identified TG as an important CAP predictor.
3. **Insulin/C-peptide** — Check whether insulin or C-peptide variables are available and whether they improve CAP prediction.
4. **Metabolic markers** — Test glucose, HbA1c and other relevant metabolic variables individually or together.
5. **BMI vs waist** — Compare BMI-only, waist-only, and BMI + waist models.
6. **Feature selection** — Apply correlation screening followed by feature selection and retrain XGBoost.
7. **Linear vs XGBoost** — Build a simple linear CAP model and compare it with your nonlinear XGBoost model.
8. **Reduced vs full features** — Test whether a smaller set of strong anthropometric/metabolic predictors can maintain performance.

**Paper Title:** Automated machine learning models for nonalcoholic fatty liver disease assessed by controlled attenuation parameter from the NHANES 2017–2020

## Findings from the paper

1. **XGBoost performed best** among the six ML models evaluated for CAP-defined hepatic steatosis.
2. **BMI, waist circumference, ALT, AST and other metabolic variables** were important predictors of steatosis severity.
3. **HOMA-IR showed strong predictive potential**, with a reported AUC of 0.831 for NAFLD prediction.
4. **HbA1c was independently associated with NAFLD**, suggesting that glycemic status may provide useful predictive information.
5. **ALT can be an early surrogate marker** for NAFLD; previous work reported an AUC of 0.715 for ALT.
6. **Feature selection was important** — the study used LASSO before training the ML models.
7. **Nonlinear relationships were important**, motivating the use of models such as XGBoost rather than only linear models.
8. **SHAP, partial-dependence plots and variable importance** were used to understand which variables drove predictions.
9. **The study cautioned about generalizability**, because training and validation were based on the same NHANES source.
10. **The authors recommend external validation and refinement across populations** before broader application.

## Potential experiments for HepatoTwin

1. **HOMA-IR** — Calculate HOMA-IR if glucose and insulin data are available and test whether it improves CAP prediction.
2. **HbA1c** — Test whether adding HbA1c provides predictive information beyond the current biochemical variables.
3. **ALT/AST** — Evaluate ALT and AST individually and together, while considering their correlation.
4. **Metabolic expansion** — Systematically add other available metabolic/liver markers and measure their effect on CV $R^2$.
5. **LASSO + XGBoost** — Use LASSO to reduce the feature set, then train XGBoost on the selected variables.
6. **Nonlinear features** — Test meaningful ratios or interactions that may capture metabolic relationships better than individual variables.
7. **SHAP/PDP analysis** — Examine whether important predictors have nonlinear effects or interactions that suggest useful feature engineering.
8. **Full vs reduced model** — Compare the current model against a smaller clinically motivated feature set.

**Paper Title:** Machine Learning Models to Predict Metabolic Dysfunction-Associated Steatotic Liver Disease (MASLD) With Simple Anthropometric and Biochemical Variables: A Cross-Sectional Study in US Population

**Relevant findings from Du et al. (2026):**

1. They began with 32 anthropometric, clinical and biochemical variables but reduced them using LASSO + Boruta rather than retaining all predictors.
2. They explicitly handled multicollinearity; for example, BMI was excluded because of its collinearity with weight and height.
3. When two predictors were closely related, they retained one and repeated feature selection.
4. Their final predictors were only weight, height, waist circumference, ALT, AST, GGT and diabetes.
5. SHAP showed waist circumference, ALT, weight, height and GGT among the most influential predictors.
6. Their XGBoost model used max depth = 4 to limit overfitting.
7. They compared several model families rather than assuming XGBoost was optimal.
8. Their task is classification rather than continuous CAP regression, so their AUC values aren't directly comparable with our $R^2$.

**Potential experiments for our model based on this paper:**

- remove BMI while retaining weight + height;
- remove redundant dietary variables;
- compare raw ALT/AST/GGT against engineered ratios;
- compare a small literature-supported feature set against our 46-feature model;
- test whether FLI/WHtR actually improve CV $R^2$;
- compare XGBoost with other tree-based regressors;
- perform a CAP=400 sensitivity analysis;
- perform an age ≥20 sensitivity analysis.

**Paper Title:** Machine Learning–Based Biomarker Identification for Early Diagnosis of Metabolic Dysfunction–Associated Steatotic Liver Disease

**Relevant findings from Boullion et al. (2025):**

1. **They used the same NHANES 2017–2020 dataset family and CAP/FibroScan measurements**, making the feature-selection methodology particularly relevant to our work. Their initial dataset had 15,560 participants and 12,471 remained after excluding hepatitis B, hepatitis C, and frequent alcohol use.
2. **They started with a broad set of demographic, metabolic, lipid, and biochemical variables**, including BMI, HbA1c, glucose, insulin, cholesterol, HDL, LDL,

triglycerides, ALT, AST, ALP, albumin, GGT, LDH, iron, bilirubin, total protein, uric acid, BUN and hs-CRP.

3. **Random Forest was used for feature-importance-based variable selection**, rather than simply putting every available feature into the final model. They also performed sensitivity analyses using **XGBoost and Recursive Feature Elimination (RFE)**.
4. **They explicitly checked multicollinearity using VIF.** This is very relevant to Harshika's suggestion to remove highly correlated variables.
5. Their VIF analysis found **extremely high multicollinearity among lipid variables**:
    - Triglycerides: **34.761**
    - HDL: **52.570**
    - LDL: **120.315**
    - Total cholesterol: **138.545**
6. **However, they did not automatically remove all correlated lipid variables.** They explain that retaining individual lipid variables can help reveal their separate relationships with steatosis, although reducing variables may improve interpretability and reliability.
7. **This is directly relevant to our current experiment:** correlation alone should not automatically determine which feature gets removed. We should test the reduced feature set using the same training CV and compare performance.
8. Their steatosis model retained **BMI, HbA1c, glucose, insulin, LDL, HDL, total cholesterol, triglycerides, ALT, AST, ALP, albumin, BUN, GGT, iron, LDH, bilirubin, uric acid and hs-CRP**, along with demographic variables.
9. Several variables showed significant differences between steatosis and normal groups, particularly **BMI, insulin, triglycerides, ALT, GGT and uric acid**.
10. **They tested different feature-selection methods and obtained very similar performance**, which is useful evidence for doing robustness checks rather than relying on one feature-selection technique. Random Forest, XGBoost and RFE produced similar steatosis-model performance.
11. Their study also emphasizes that **CAP/FibroScan itself has measurement limitations** and that CAP does not perfectly correspond to histological liver disease.
12. **Their task is classification, not continuous CAP regression.** Their reported accuracy/sensitivity/specificity therefore should **not be directly compared with our MAE/RMSE/$R^2$**. Their methodological choices are what matter for our model

**Paper Title-** Machine learning-based prediction of metabolic dysfunction-associated steatotic liver disease using National Health and Nutrition Examination Survey (NHANES) data

**Relevant findings from Zhang et al. (2025):**

1. **They started with 24 candidate features but reduced them to 12 using RFE.** The purpose was explicitly to reduce overfitting caused by feature redundancy.
2. **Feature reduction barely changed performance.** Their initial XGBoost model had test AUC **0.8738**, while the 12-feature RFE model had **0.8740**. This is strong evidence that many of the original features were unnecessary for their prediction task.

3. Their final 12 features were:
    - Age
    - Race
    - BMI
    - Waist circumference
    - ALT
    - BUN
    - Globulin
    - GGT
    - HDL
    - Triglycerides
    - CRP
    - Glucose
4. **Waist circumference was the strongest feature according to SHAP**, followed by glucose and triglycerides.
5. The paper compared **five different algorithms** rather than assuming XGBoost was automatically optimal: LR, RF, LightGBM, CatBoost and XGBoost. XGBoost had the highest test AUC in their experiment.
6. Their XGBoost configuration is interestingly close to ours: **learning rate = 0.02, max depth = 4, subsample = 0.8**, with additional regularization.
7. They used **RFE iteratively**: remove the least-contributing feature, retrain, evaluate, and continue until the desired feature count/performance criterion is reached.
8. Several of their candidate variables are directly relevant to our model: **BMI, waist, ALT, AST, GGT, glucose, triglycerides, HDL and metabolic variables**.
9. Their SHAP results support the importance of **anthropometric + metabolic variables**, particularly waist circumference, glucose and triglycerides.
10. **Important limitation:** this is a **MASLD classification problem**, not continuous CAP regression. Their AUC of 0.874 therefore cannot be directly compared with our $R^2$. We should use the paper mainly to motivate **feature selection, model comparison and ablation experiments**. Their target uses CAP ≥302 dB/m as the fatty-liver classification criterion.